

Delyan Boychev

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Sofia, Bulgaria

Research Interests

Efficient and multimodal deep learning, with a focus on:

- Hardware-aware algorithms and GPU kernels for fast training and inference.
- Vision-language models for remote sensing and 3D scene processing.

Education

INSAIT Scholar (highly selective 4-year program) Sofia, Bulgaria
Faculty of Mathematics and Informatics, Sofia University
Bachelor of Informatics *Sept. 2024 – Expected 2028*

Experience

Undergrad Research Scholar | INSAIT Oct. 2024 – Present

- **Efficient training and inference** (advised by Dr. Dimitar I. Dimitrov): accelerating relative-position attention with exact, fused Triton kernels, using offset-aware tiling, cross-tile reuse, and an atomic-free backward pass [1].
- At 8K tokens: up to 6.4× faster forward, 21× faster backward, and 128× lower peak memory than the strongest baselines; 1.8× faster end-to-end speech transcription.
- **Remote-sensing VLMs** (advised by Dr. Danda Paudel): adapting vision-language models to satellite imagery with OpenStreetMap-derived supervision (OSMDA [2]).
- **3D scene processing** (advised by Dr. Danda Paudel): data-efficient structured scene understanding, building on SpatialLM with a Chorus-based 3D Gaussian encoder to lower training-data requirements (ongoing).

NLP Research Intern | Graphwise July 2026 – Sept. 2026
Mentors: Dr. Petar Mitankin, Dr. Nikola Tulechki, Dr. Philip Ganchev, Zlatina Marinova

- Built NLP pipelines for table entity reconciliation, knowledge-base linking, and structured extraction from unstructured text.
- Optimized language-model pipelines for deployment, improving inference cost and latency.

Quant ML Intern | Man Group July 2025 – Sept. 2025
Mentors: Dr. Yavor Kovachev, Dr. Diana Zheng

- Conducted Quant ML research, focusing on developing and evaluating machine-learning models for systematic trading and signal forecasting.

Publications

[1] **Fast and Exact Relative-Position Attention via Structure-Aware Kernels**
Delyan Boychev, Ivo Petrov, Dimitar I. Dimitrov. *arXiv preprint, 2026. Under review.* arXiv:2611.01182

[2] **OSMDA: OpenStreetMap-based Domain Adaptation for Remote Sensing VLMs**
Stefan Maria Ailuro, Mario Markov, Mohammad Mahdi, **Delyan Boychev**, Luc Van Gool, Danda Pani Paudel. *arXiv preprint, 2026. Under review.* arXiv:2603.11804

[3] **ImagiNet: A Multi-Content Benchmark for Synthetic Image Detection**
Delyan Boychev, Radostin Cholakov. *AAAI 2025 Workshop on Datasets and Evaluators of AI Safety.* OpenReview | arXiv:2407.20020

[4] **Interpretable Computer Vision Models through Adversarial Training**
Delyan Boychev. *arXiv preprint, 2023.* arXiv:2307.02500

Honors

- Scholarship Recipient, EXPLORER Program, INSAIT (2024)
- European Space Agency Award from EUCYS'24, Katowice, Poland
- International Olympiad in Artificial Intelligence 2024, Team Bronze Medal
- Regeneron ISEF'23 Finalist, Dallas, Texas
- John Atanasoff Presidential Award in the category "Debut breakthrough in computer technology" (2023)

Projects

DisentangledFlash | *Triton Attention Kernel* Aug. 2026 – Present

- Fuses DeBERTa-v2/v3 attention into a streaming Triton kernel without materializing the full attention matrix.
- Benchmarked at up to 5.18× faster forward and 2.15× faster forward+backward than Hugging Face Transformers eager.
- Integrated into GLiNER2 (fastino-ai) as an optional attention backend, released in v2.1.0. Also contributed span-scoring and span-decoding speedups (up to 1.92×) to GLiNER2.
- The paper [1] is a continuation of this project.
- **Code:** github.com/delyan-boychev/disentangled-flash | **PyPI:** pypi.org/project/disentangled-flash

LavBench | *ML Competition Platform* June 2026 – Present

- A secure, bilingual platform for machine-learning competitions; the official evaluation system of the Bulgarian National Olympiad in AI.
- Runs notebook and Python submissions in hardened Docker sandboxes; provides live leaderboards, double-blind jury review, custom evaluators, and distributed workers.
- **Code:** github.com/delyan-boychev/lavbench | **Docs:** lavbench.readthedocs.io

ImagiNet | *Synthetic Image Dataset* 2024

- A 144 GB, 200K-image dataset for synthetic-image detection, balanced across photos, paintings, faces, and uncategorized imagery.
- Supports both real-versus-synthetic classification and generator identification across open and proprietary image generators.
- **Dataset:** huggingface.co/datasets/delyanboychev/imaginet

Gradient Cache Contrastive Learning | *PyTorch Package* 2023 – 2024

- Implemented gradient caching in PyTorch to scale contrastive learning beyond GPU memory limits.
- Reproduces large-batch gradients from smaller chunks; supports SimCLR, SupCon, SelfCon, mixed precision, and distributed training.
- **Code:** github.com/delyan-boychev/grad-cache-con-learning | **PyPI:** pypi.org/project/grad-cache-con-learning

Teaching & Service

National Committee Member & Lecturer | Bulgarian National Olympiad in AI June 2025 – Present

- Member of the national committee organizing the Bulgarian National Olympiad in AI: writing competition tasks and running contests on LavBench (see Projects).
- Lecturing and mentoring students preparing for the national olympiad and the International Olympiad in Artificial Intelligence (IOAI).

Technical Skills

- **Programming:** Python, C++, TypeScript, JavaScript
- **ML/Research:** PyTorch, Triton, vLLM, Hugging Face Transformers
- **Data Science:** NumPy, pandas, scikit-learn
- **Web & Backend:** React, Vite, Tailwind CSS, Flask, Celery, PostgreSQL
- **Tools:** Linux, Docker, Google Cloud, CI/CD, Git

Languages

- Bulgarian — Native
- English — IELTS Academic, 7.0/9.0